INDIA INNOVATES 2026

Team Name:
NavaBharataya



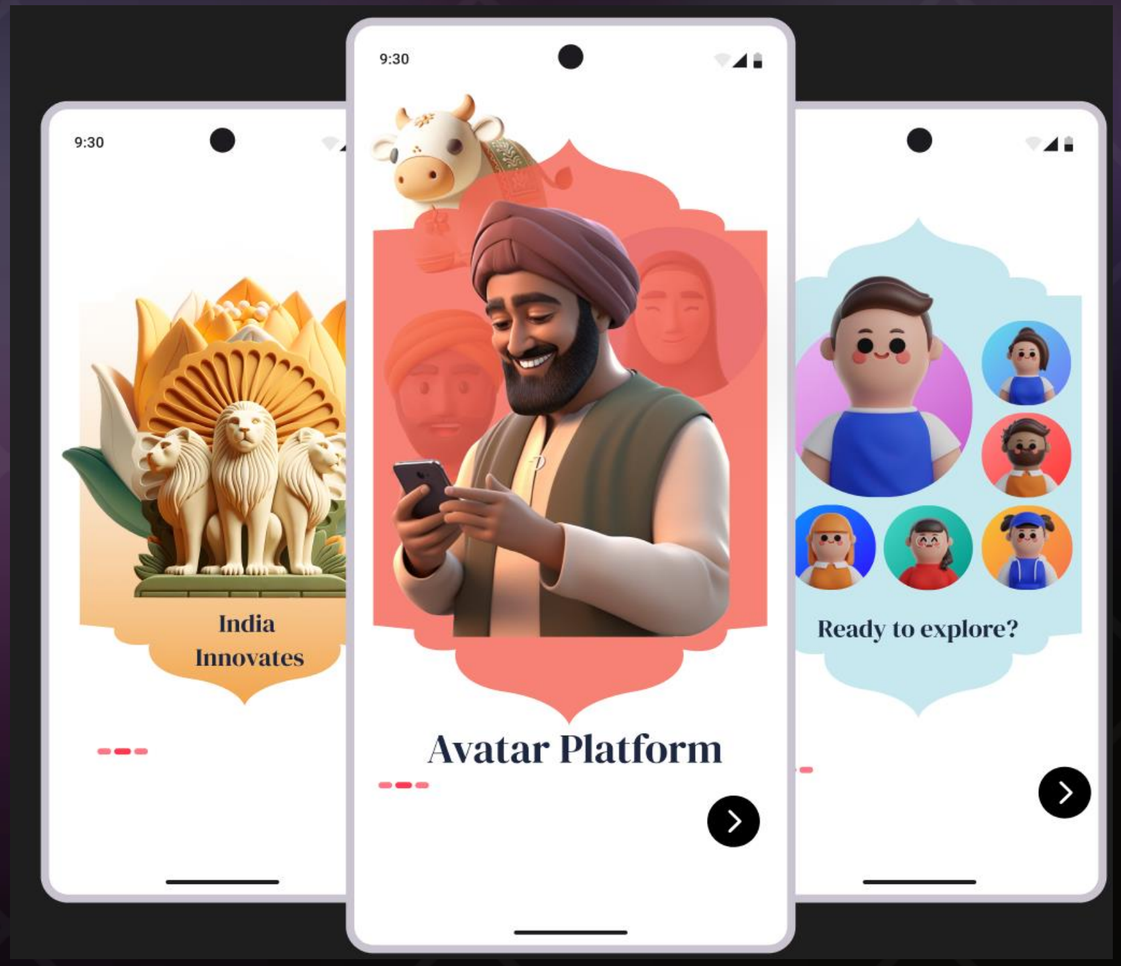
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PROBLEM STATEMENT

Given Statement:
Designing and building a platform that enables individuals, leaders, educators, institutions, and organizations to create realistic, interactive digital avatars capable of speaking, presenting, and engaging audiences in real time across multiple languages for governance, education, outreach, and large-scale communication explaining the work completed and civic impact in a clear and transparent manner.

Explanation:
People usually suffer from lack of information about their government schemes and the relevant information of their respective local governance and national governance. They do not get support in their native languages and proper explanation. They also cannot opinionate about the schemes and other relevant topics. This causes a major gap between the people, businesses and the government.



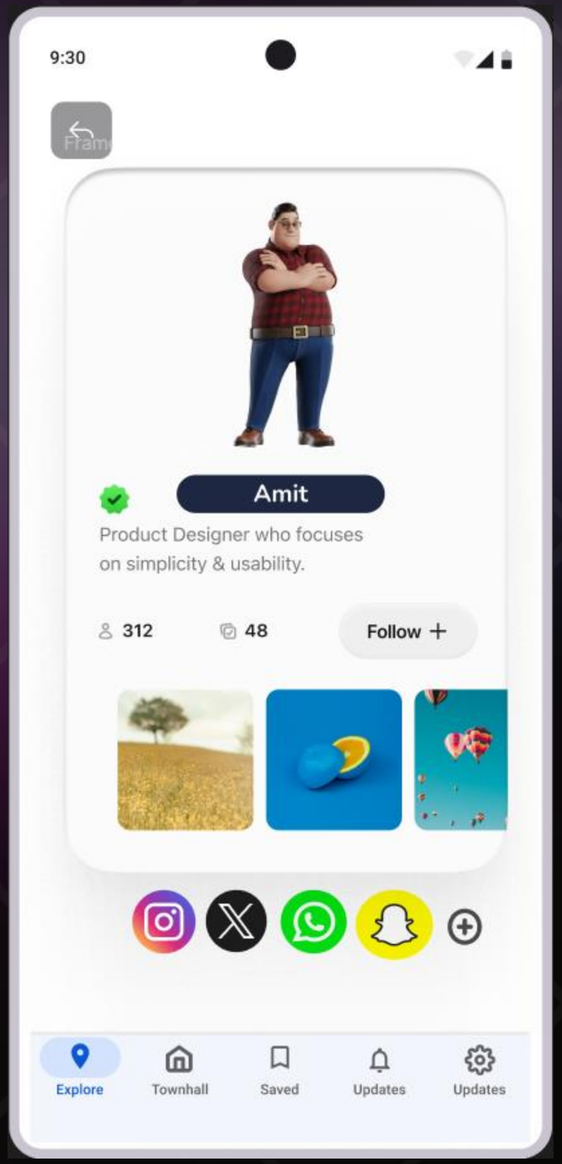
Prototype Preview

SOLUTION

We propose an AI-powered avatar platform that enables verified authorities and institutions to communicate directly with citizens through interactive digital avatars. These avatars provide real-time, multilingual explanations of government schemes, infrastructure projects, and policies, helping bridge the communication gap between citizens and governance. The system combines secure identity verification, AI-driven knowledge retrieval, and location-based information delivery to ensure transparent and accessible civic engagement.

Key Features:

- Verified Digital Avatars: Secure 3D avatars for leaders and authorities, and lightweight 2D avatars for citizens to enable trusted communication.
- Multilingual AI Interaction: Avatars interact with users in multiple Indian languages using AI translation and speech systems.
- Smart Complaint Handling: Citizens can submit text, images, or videos of issues; the system routes them to the relevant authority avatar for explanation and action.
- Geo-Fencing Civic Awareness: Location-based triggers provide nearby citizens with information about ongoing infrastructure and government projects.
- AI Townhalls: Virtual townhall sessions where citizens can interact with authority avatars, ask questions, and receive policy explanations in real time.



ARCHITECTURE

System Architecture

The platform follows a modular AI-driven architecture that enables users to create digital avatars, interact with AI systems, and access real-time civic information. The system integrates secure authentication, avatar generation, multilingual AI processing, government knowledge retrieval, and geo-fencing services to provide transparent and interactive communication between citizens and authorities.

Architecture Workflow

- User Authentication: Users securely log in and are classified as civilians or verified authorities.
- Avatar Creation Engine: Civilians generate 2D AR-based avatars, while authorities create secure 3D avatars for official communication.
- Avatar Dashboard: Users manage interactions, submit complaints, explore maps, and communicate with other avatars.
- AI Interaction Engine: The system uses Sarvam AI and LLMs to enable multilingual conversations and intelligent responses.
- Knowledge & RAG System: AI retrieves verified information from government sources, notices, and real-time updates to answer queries.
- Geo-Fencing & Civic Awareness: Map-based system detects nearby development projects and triggers context-aware notifications and civic impact explanations.
- End-to-End Avatar Communication: The avatar processes queries and delivers real-time, multilingual responses to users.

TECHNOLOGY USED

Frontend & User Interface
- React / Next.js – Builds the interactive web interface and avatar dashboard.
- React Native (Mobile App) – Enables cross-platform mobile access for citizens and authorities.
- Mapbox / Google Maps API – Provides geolocation services and geo-fencing map visualization.

Backend & APIs
- Node.js / FastAPI – Handles backend logic, API routing, and system communication.
- REST APIs & WebSockets – Enable real-time data exchange and interactive avatar communication.

Artificial Intelligence & NLP
- Sarvam AI – Multilingual processing for speech recognition, translation, and voice synthesis.
- Large Language Models (LLMs) – Power conversational intelligence and automated responses.
- LangChain / LlamaIndex – Implement Retrieval Augmented Generation (RAG) pipelines.

Data & Knowledge Systems
- Vector Database (Pinecone / Weaviate) – Stores embeddings for fast retrieval of government data.
- PostgreSQL / NoSQL Databases – Manages user profiles, complaints, and system data.

Avatar & Media Generation
- Unreal Engine / MetaHuman or Ready Player Me – Generates realistic 3D authority avatars.
- AR / WebGL Technologies – Supports lightweight 2D avatars for civilians.

Cloud & Infrastructure
- AWS / Google Cloud Platform – Scalable hosting, storage, and computing resources.
- Docker & Kubernetes – Containerization and orchestration for system scalability.

FEATURE / USP

Key features:

1. Multilingual Communication
   AI-Powered speech and translations to allow citizens to interact with avatars and gain information in their native language.

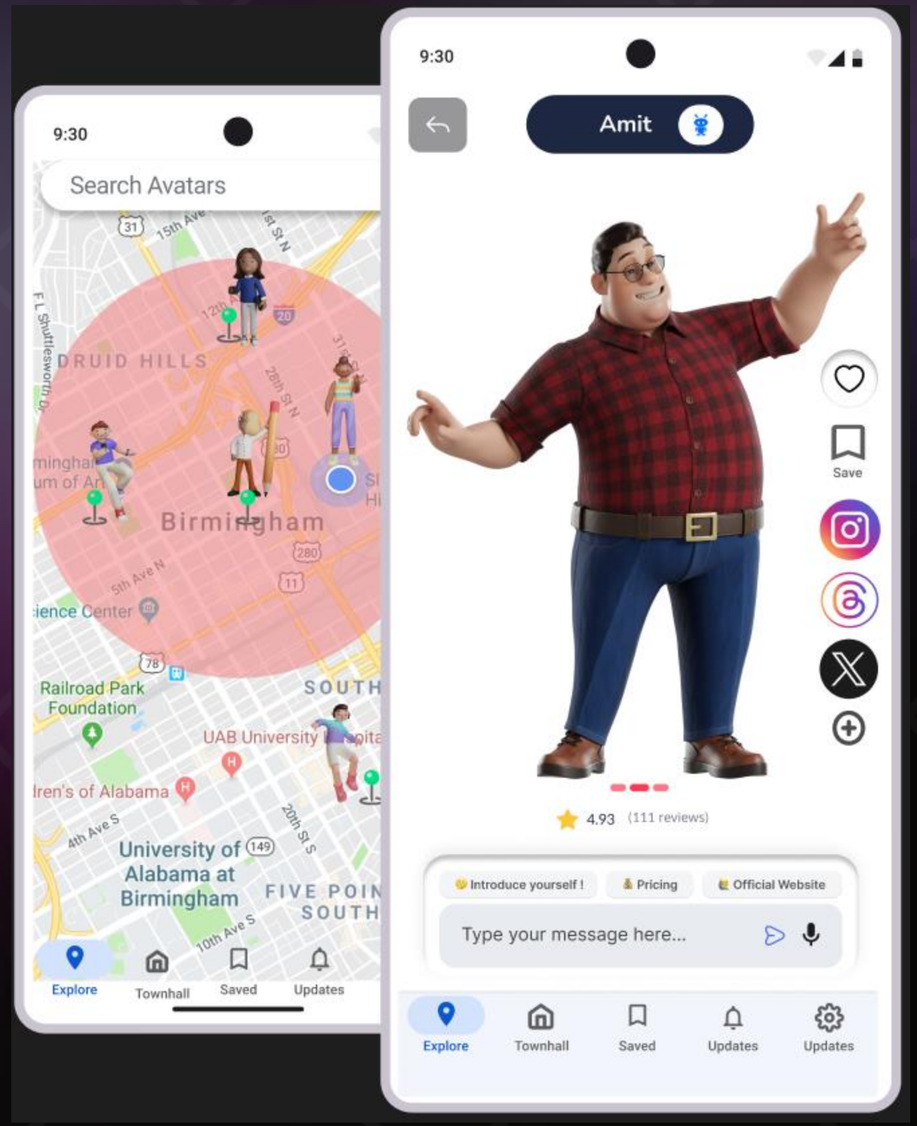
2. AI-Powered Digital Townhalls
   Interactive virtual sessions where citizens can ask and raise questions to the authorities and receive transparent policy explanations about their local government and/or locality from their respective representative avatars.

3. Hyper-local Geo-Fencing awareness
   Location-based notifications explaining government projects, policies, infrastructure development and their impact.

4. Smart Avatar-Powered Representatives
   3D-Avatars that act as representatives of their respective roles that can answer queries of the public at scale accurately using the government policies' knowledge base.

5. Real-Time Governance Alerts
   Citizens get alerts about the policies and actions of their local governments where they can participate and opinionate.

6. Knowledge using RAG
   Avatars provide accurate answers using verified government data, notices and real-time updates.



REFERENCES / LINKS

1. Sarvam AI: [https://docs.sarvam.ai/sarvam-api-docs]
2. OpenStreetMap: [https://taginfo.openstreetmap.org/taginfo/apidoc]
3. RAG Avatar Memory: [https://www.sciencedirect.com/science/article/pii/S2666188825006586]
4. GitHub Repository: https://github.com/Arpan010/AI_Summit.git]

THANK YOU